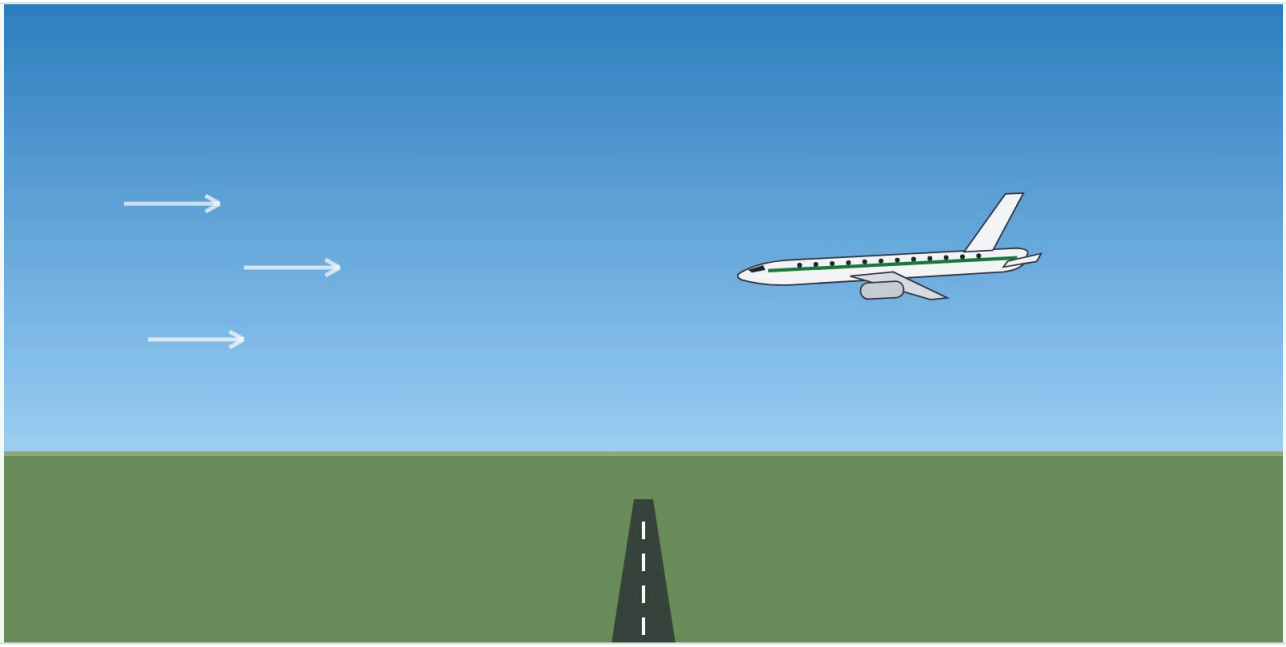
MODULE 1 · WIND AND LIMITATIONS

Lesson 01: Wind Reports and Wind Components

Lesson 1 of 3 · about 6 min

LEARNING OBJECTIVES

- Interpret wind reports correctly.
- Calculate headwind, tailwind and crosswind components.
- Use gusts correctly.

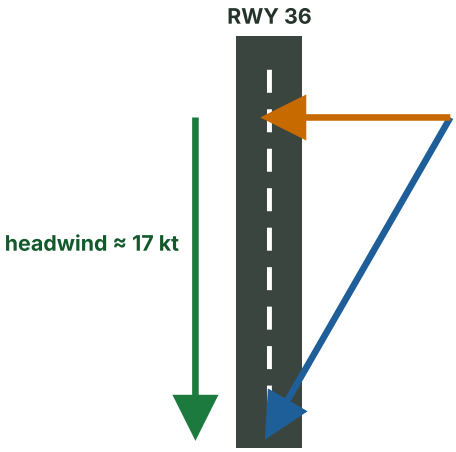


Calculate the components with the gust.

Wind reports

- In the **METAR**, the wind is a 10-minute mean; in reports for **take-off and landing** from the tower (and ATIS), it is a 2-minute mean for the runway (ICAO Annex 3).
- **Gusts** are reported when the maximum wind exceeds the mean by 10 kt or more: `27015G28KT` = 270°, 15 kt, gusting 28 kt.
- Variable direction: `27015KT 240V300` — direction varies between 240° and 300°.
- Directions in METAR / TAF are **true**; tower and ATIS winds are **magnetic**.

Components



Rule of thumb (crosswind)

30° off the runway → ½ of the wind
45° → about 0.7 of the wind
60° → about 0.9 of the wind
90° → all of the wind

crosswind = wind × sin(angle)
headwind = wind × cos(angle)

Always use the gust value when checking a crosswind limit, unless the company procedure states otherwise.

- Use the FMS / EFB, the wind component chart or the rule of thumb.
- For limits, check the crosswind with the **gust** and the tailwind with the steady wind and the gust as your procedure states.
- With a variable wind direction, use the most unfavourable direction.

KEY TAKEAWAYS

- METAR wind: 10-min mean, true; tower / ATIS: 2-min mean, magnetic.
- Gusts reported when 10 kt or more above the mean.
- Crosswind ≈ wind × sin(angle); use the gust for crosswind limits.

KNOWLEDGE CHECK

Wind 090°/20 kt on runway 06. Approximately what is the crosswind?

A. 5 kt
B. 10 kt
C. 17 kt
D. 20 kt

Answer: B. 30° off the runway: half the wind, about 10 kt.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.